

Tarun Singh

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EXPERIENCE

Data Scientist

March 2025 – Present

OMR India Outsources Pvt Ltd

Bengaluru, India

- Fine-tuned and deployed a lightweight LLM for movie script summarization and genre classification, improving genre-classification accuracy by **30%** over the base model and cutting manual script review time by about a third.
- Built a movie recommendation system for Reeloid using movie-category embeddings and cosine similarity, increasing clicks on recommended titles by **20%** and lengthening average user sessions.
- Built a LangChain-based LLM solution to convert video content into quizzes by processing video → audio → transcript → questions, cutting manual question-creation effort by **40%**.
- Developed **60+** REST APIs for the Reeloid application using FastAPI and MongoDB, and optimized **100+** existing endpoints, cutting average API response time by **50%**.

AI/ML Career Transition & Upskilling

February 2024 – February 2025

Self-Directed Study & Project Development

- Completed focused upskilling in Machine Learning, Deep Learning, NLP, and Generative AI through structured study and project development.
- Built and deployed 3 end-to-end AI/ML projects spanning RAG pipelines, transformer fine-tuning, and recommendation systems.

Quality Assurance Engineer

June 2023 – February 2024

Paytm

Delhi, India

- Implemented and automated **100+** end-to-end test cases in Java within IntelliJ IDEA, integrating suites into the CI pipeline to reduce manual testing and speed up release verification by **20%**.
- Designed and refactored a modular, data-driven test framework, improving test reusability and reducing flaky failures by **35%**, which helped catch more defects in critical user flows before release.
- Led cross-functional testing sprints with Product and Engineering to define acceptance criteria and risk-based test plans, decreasing test-case development and execution time by **29%** and shortening bug-to-fix cycles.
- Conducted automated regression and backward-compatibility testing for weekly releases and introduced selective regression runs, reducing regression escapes (bugs reaching production) by **30%**.

PROJECTS

YouTube Comments Sentiment Analysis | Source Code

- Developed a YouTube comment sentiment analyzer by fine-tuning MobileBERT on the DAIR-AI Emotion dataset (16K labeled samples) to classify comments into six emotions, reaching **91.9%** accuracy on a 2K-sample held-out test set.
- Optimized inference with TensorFlow Lite Float16 quantization, halving model size while keeping **91.1%** test accuracy.
- **Tech used:** Deep Learning, MobileBERT, Transformers, NLP, TensorFlow, TensorFlow Lite, Quantization, YouTube API.

Harry Potter In Depth Query Bot | Source Code

- Built a Harry Potter query bot using LangChain RAG with Chroma vector search over all seven books (**1M+** words).
- Implemented embeddings and prompt templates to reduce hallucinations; deployed serverless on AWS Lambda.
- **Tech used:** Generative AI, GPT-4o-mini, LangChain, Chroma, RAG, Embeddings, Prompt Engineering, Flask, Docker

Bollywood Song Recommendation System | Source Code

- Built a Bollywood song recommender using Spotify metadata and cosine similarity across a **1,445**-song catalog.
- **Tech used:** Machine Learning, Python, Scikit-learn, Flask, Docker, DVC, Git.

SKILLS

Languages & Libraries : Python, TensorFlow, PyTorch, Scikit-Learn, Hugging Face, TRL, LoRA, QLoRA, PEFT, NumPy, Pandas

Tools : AWS, Docker, DVC, MLflow, ONNX, MLOps, Git / GitHub, Pruning, Knowledge Distillation, Edge Deployment

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

August 2019 - June 2023

Bachelor of Technology in Computer Science and Engineering

Uttar Pradesh, India